

SELF-SUSTAINING SOLAR BUOY FOR OCEANIC WEATHER AND POLLUTION ANALYSIS

Major Project Phase II Report

submitted by

**Mr. TASMAY SHETTY
(NNM22EC188)**

**Ms. VAISHNAVI S
(NNM22EC195)**

**Ms. VARSHA R SHETTY
(NNM22EC197)**

**Mr. JAYANTH P R
(NNM23EC511)**

To



(Estd. under Section 3, UGC Act 1956)

(Placed under Category 'A' by MHRD, Govt. of India; Accredited as 'A+' Grade University by NAAC)

In partial fulfilment of the requirements for the award of

**BACHELOR OF TECHNOLOGY IN ELECTRONICS AND
COMMUNICATION ENGINEERING**

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

NMAM INSTITUTE OF TECHNOLOGY, NITTE

April 2026

SELF-SUSTAINING SOLAR BUOY FOR OCEANIC WEATHER AND POLLUTION ANALYSIS

Major Project Phase II Report

submitted by

**Mr. TASMAY SHETTY
(NNM22EC188)**

**Ms. VAISHNAVI S
(NNM22EC195)**

**Ms. VARSHA R SHETTY
(NNM22EC197)**

**Mr. JAYANTH P R
(NNM23EC511)**

Under the Guidance of

**Dr. K. V. S. S. S. SAIRAM
Professor & Head**

To



(Estd. under Section 3, UGC Act 1956)

(Placed under Category 'A' by MHRD, Govt. of India; Accredited as 'A+' Grade University by NAAC)

In partial fulfilment of the requirements for the award of

**BACHELOR OF TECHNOLOGY IN ELECTRONICS AND
COMMUNICATION ENGINEERING**

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

NMAM INSTITUTE OF TECHNOLOGY, NITTE

April 2026

Department of Electronics and Communication Engineering**CERTIFICATE**

This is to certify that the Project Report of the Project Work titled “**Self-Sustaining Solar Buoy for Oceanic Weather and Pollution Analysis**” is a bonafide research work carried out by **Tasmay Shetty (NNM22EC188), Vaishnavi S (NNM22EC195), Varsha R Shetty (NNM22EC197), Jayanth P R (NNM23EC511)** under the guidance of **Dr. K. V. S. S. S. Sairam** and co-guidance of **Mr. Chandra Singh** in the **Department of Electronics and Communication , NMAM Institute of Technology, Nitte.**

The same is being submitted to the Nitte (Deemed to be University) in partial fulfillment of the requirements for the award of **Bachelor of Technology in Electronics and Communication Engineering.**

Name & Signature of Guide(s)**Name & Signature of HOD****Head of the Institution****Project Viva voice****Name of the Examiners****Signature with Date**

1. _____

2. _____

ACKNOWLEDGEMENT

We express our sincere gratitude to our respected **Dr. Niranjan N. Chiplunkar, Principal of NMAM Institute of Technology** for providing us with the opportunity and necessary facilities to carry out this project successfully in the institution. His constant support and encouragement have played a vital role in the completion of this work.

We express our heartfelt thanks to our project guide **Dr. K. V. S. S. S. Sairam, Professor and Head**, Department of Electronics and Communication Engineering, NMAM Institute of Technology, for his valuable guidance, continuous support, and encouragement throughout the project. His technical expertise and insightful suggestions have greatly contributed to the successful completion of this work.

We would also like to express our sincere thanks to our co-guide **Mr. Chandra Singh** for his guidance, support, and valuable suggestions during the course of this project. His inputs have been highly beneficial in improving the quality of our work.

We also extend our sincere thanks to **Dr. Charishma, Assistant Professor (Grade III)** and **Mr. Chandra Singh, Assistant Professor (Grade II)** Project Head, Department of Electronics and Communication Engineering, NMAM Institute of Technology, Nitte, for her guidance, support, and encouragement throughout the project work.

We would like to thank all the faculty members of the Department of Electronics and Communication Engineering for their support and for providing the necessary resources to carry out this project. Their guidance has helped us enhance our Skills.

ABSTRACT

The project titled “*Self-Sustaining Solar Buoy for Oceanic Weather and Pollution Analysis*” aims to develop an autonomous IoT-based buoy for monitoring real-time water quality and environmental parameters in aquatic environments. The system integrates multiple sensors to measure key parameters, providing valuable insights into surrounding conditions. An Arduino Nano serves as the primary processing unit, collecting and organizing sensor data, which is transmitted wirelessly using the NRF24L01 module to an ESP32 microcontroller. The ESP32 uploads the data to the Blynk cloud platform, enabling real-time monitoring and historical data visualization through a user-friendly interface. A Python-based machine learning model is implemented to analyze sensor data and predict the likelihood of algae bloom formation, enabling early detection of environmental hazards. The system is powered by solar energy, ensuring continuous and maintenance-free operation even in remote locations while reducing dependence on external power sources. Furthermore, the system is designed to be cost-effective, scalable, and energy-efficient, making it suitable for deployment in oceans, lakes, and rivers. By integrating IoT, wireless communication, renewable energy, and intelligent data analysis, the system provides an efficient solution for environmental monitoring, pollution detection, and ecosystem management.

LIST OF FIGURES

Figure No.	Title	Page No.
Fig. 3.1	Block Diagram of Self-Sustaining Solar Buoy for Oceanic Weather and Pollution Analysis	8
Fig. 4.1	Zero PCB Board	15
Fig. 4.2	Arduino Nano Microcontroller	17
Fig. 4.3	LM7805 Voltage Regulator	19
Fig. 4.4	LM7833 3.3V Voltage Regulator	20
Fig. 4.5	nRF24L01 RF Transceiver Module	22
Fig. 4.6	Solar Panel	24
Fig. 4.7	Rechargeable Battery	26
Fig. 4.8	Battery Management System	28
Fig. 4.9	MPU6050 Sensor Module	29
Fig. 4.10	DHT22 Sensor	31
Fig. 4.11	pH Sensor Module	32
Fig. 4.12	Turbidity Sensor Module	34
Fig. 4.13	ESP32 Microcontroller	36
Fig. 4.14	LCD Display	38
Fig. 5.1	Arduino IDE Interface	40
Fig. 5.2	Blynk Dashboard Interface	40
Fig. 5.3	Google Colaboratory	41
Fig. 6.1	Final Prototype of Water Buoy System	42
Fig. 6.2	Blynk Dashboard Output	43
Fig. 6.3	Transmitter Unit Setup	44
Fig. 6.4	Receiver Unit Setup	45

NOMENCLATURE

Symbol / Abbreviation	Description
Arduino Nano	Microcontroller used for data acquisition and processing in the transmitter unit
ESP32	Microcontroller used for Wi-Fi communication and cloud data uploading
nRF24L01	RF transceiver module used for wireless data transmission between transmitter and receiver
LM7805	5V voltage regulator used to provide stable power supply to electronic components
LM7833	3.3V voltage regulator used for low-voltage modules such as RF communication devices
Solar Panel	Device used to convert solar energy into electrical energy for powering the system
Rechargeable Battery	Energy storage device used to supply power during low or no sunlight conditions
BMS	Battery Management System used to protect and manage battery charging and discharging
MPU6050	Sensor module used to measure motion, acceleration, and orientation of the buoy
DHT22	Digital sensor used for measuring environmental temperature and humidity
pH Sensor	Sensor used to measure acidity or alkalinity level of water
Turbidity Sensor	Sensor used to measure water clarity by detecting suspended particles
LCD	Liquid Crystal Display used for real-time local display of sensor data
IoT	Internet of Things system used for remote monitoring and data visualization
RF Communication	Wireless communication method used for data transfer using radio frequency
Voltage Regulation	Process of maintaining constant voltage levels for safe operation of components
Cloud Platform	Online system used for storing, processing, and visualizing sensor data
Transmitter Unit	Section of system responsible for sensing and sending environmental data
Receiver Unit	Section of system that receives data and uploads it to cloud
Embedded System	Combination of hardware and software designed for specific monitoring application
Real-time Monitoring	Continuous observation and reporting of environmental parameters
Environmental Parameters	Measured variables such as temperature, humidity, pH, turbidity, and motion
Renewable Energy	Energy derived from solar power used to operate the system sustainably

TABLE OF CONTENT

CERTIFICATE.....	i
ACKNOWLEDGEMENT.....	ii
ABSTRACT.....	iii
LIST OF FIGURES.....	iv
NOMENCLATURE.....	v
TABLE OF CONTENTS.....	vi
CHAPTER 1: INTRODUCTION.....	1
1.1 Background.....	1
1.2 Problem Statement.....	2
1.3 Objectives.....	2
1.4 Scope of the Project.....	3
CHAPTER 2: LITERATURE REVIEW.....	4
2.1 Existing Systems.....	4
2.2 Limitations of Existing Systems.....	5
2.3 Proposed System Overview.....	6
CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY.....	8
3.1 Block Diagram.....	8
3.2 Working Principle	10
3.3 Algorithm.....	10
CHAPTER 4: HARDWARE REQUIREMENTS.....	15
4.1 Hardware Requirements.....	15
4.1.1 Zero PCB	
4.1.2 Arduino Nano Microcontroller	
4.1.3 LM7805 Voltage Regulator (5V)	
4.1.4 LM1117 3.3V Voltage Regulator	
4.1.5 nRF24L01 RF Transceiver Module	
4.1.6 Solar Panel	
4.1.7 Rechargeable Battery	
4.1.8 Battery Management System (BMS)	
4.1.9 MPU6050 Sensor Module	
4.1.10 DHT22 Temperature and Humidity Sensor	
4.1.11 pH Sensor Module	
4.1.12 Turbidity Sensor Module	
4.1.13 ESP32 Microcontroller	
4.1.14 LCD Display	

CHAPTER 5: SOFTWARE REQUIREMENTS.....	40
5.1 Arduino IDE	
5.2 Blynk IoT Platform	
5.3 Google Colaboratory	
CHAPTER 6: RESULTS AND DISCUSSION	42
6.1 Observations	42
6.2 Performance Analysis	44
6.3 Advantages	46
6.4 Discussion of Findings.....	46
6.5 Performance Analysis and Challenges.....	47
6.6 Environmental and Economic Impact.....	48
CHAPTER 7: APPLICATIONS	49
CHAPTER 8: FUTURE SCOPE	50
CHAPTER 9: CONCLUSION	51
REFERENCES	52

CHAPTER 1

INTRODUCTION

1.1 Background

Monitoring marine and freshwater environments is essential for understanding weather patterns, detecting pollution, and preserving aquatic ecosystems. Conventional methods of water quality assessment often involve manual sampling, which is time-consuming, costly, and lacks real-time capability. With growing environmental concerns such as climate change, marine pollution, and harmful algal blooms, there is a pressing need for an automated, continuous, and reliable monitoring solution that can operate independently in harsh conditions.

The Self-Sustaining Solar Buoy for Oceanic Weather and Pollution Analysis is designed to address these challenges by integrating IoT-based sensing, solar power, and machine learning. The buoy is equipped with sensors such as the MPU6050 for stability detection, DHT22 for temperature and humidity measurement, and a turbidity sensor for water clarity monitoring. The data collected is transmitted from the Arduino Nano via NRF24L01 to an ESP32, which uploads it to the Blynk cloud for real-time and historical visualization. A Python-based ML model then processes the data to predict the likelihood of algae blooms, allowing early intervention in pollution and ecosystem management.

By combining renewable energy, wireless communication, and intelligent analytics, this project demonstrates a sustainable approach to environmental monitoring. The self-powered design ensures continuous operation without external energy sources, while the integration of machine learning enhances prediction accuracy and decision-making. Ultimately, the system offers a scalable, cost-effective, and eco-friendly solution for long term water quality and weather monitoring, contributing to the protection and preservation of aquatic environments.

1.2 Problem Statement

Conventional water quality monitoring systems are limited by high operational costs, lack of real-time data access, and dependence on manual intervention. These methods are not suitable for continuous monitoring, especially in remote or offshore locations where accessibility is difficult. Additionally, existing monitoring solutions often lack predictive capabilities, making it challenging to identify potential environmental hazards such as pollution spikes or algal bloom formation in advance.

There is a need for an efficient, cost-effective, and autonomous system that can continuously monitor environmental parameters and provide real-time data along with predictive analysis. Such a system should be capable of operating independently with minimal maintenance while ensuring reliable performance in harsh environmental conditions.

1.3 Objectives

The main objectives of this project are as follows:

- To design and develop an autonomous IoT-based buoy system for real-time environmental monitoring.
- To integrate sensors for measuring water quality and atmospheric parameters.
- To implement wireless communication for efficient data transmission between transmitter and receiver units.
- To enable real-time data visualization using a cloud-based platform.
- To develop a machine learning model for predicting potential environmental risks such as algal blooms.
- To ensure continuous operation using solar energy for a self-sustaining system.
- To design a cost-effective and scalable solution suitable for large-scale deployment.

1.4 Scope of the Project

The scope of this project includes the design, development, and implementation of a self-sustaining solar-powered buoy system capable of monitoring environmental and water quality parameters in real time. The system focuses on integrating sensing, data processing, wireless communication, and cloud-based visualization into a single platform.

The project is intended for applications such as water pollution monitoring, marine research, aquaculture management, and environmental studies. It is designed to operate in remote aquatic environments with minimal human intervention. While the current system provides real-time monitoring and basic predictive analysis, further enhancements such as advanced sensors, improved communication technologies, and large-scale deployment can expand its capabilities in future applications.

CHAPTER 2

LITERATURE REVIEW

2.1 Existing Systems

Monitoring of aquatic environments has gained significant importance due to increasing concerns over water pollution, climate change, and ecosystem degradation. Traditional monitoring systems primarily rely on manual sampling and laboratory analysis, which are time-consuming, labor-intensive, and lack real-time capabilities [1]. These methods also suffer from limited spatial and temporal coverage, making them unsuitable for continuous environmental assessment.

With the advancement of technology, researchers have focused on developing automated monitoring systems using the Internet of Things (IoT), wireless sensor networks, and embedded systems [2]. These systems utilize multiple sensors to measure environmental parameters and transmit the data to remote servers for analysis. IoT-based monitoring systems enable real-time data acquisition and remote accessibility, significantly improving the efficiency of environmental monitoring.

Several existing systems are based on buoy platforms equipped with sensors for measuring water quality and atmospheric conditions. These buoy-based systems are widely used in marine and freshwater environments for continuous monitoring. Wireless communication technologies such as RF modules, Wi-Fi, and GSM are commonly employed to transmit collected data to cloud platforms, enabling real-time visualization and analysis [4]. Such systems reduce the need for human intervention and improve monitoring efficiency.

Recent studies have also explored the integration of artificial intelligence (AI) and machine learning (ML) techniques in environmental monitoring systems. These technologies are used to analyze large datasets, identify patterns, and predict environmental changes. For instance, AI-based buoy systems have been developed for aquaculture monitoring, where they analyze water parameters to optimize fish farming conditions [2]. These systems demonstrate the potential of intelligent monitoring solutions in improving environmental management.

In addition to IoT-based systems, remote sensing technologies have been widely used for large-scale environmental monitoring. Satellite-based observations provide extensive coverage of oceans and water bodies, enabling the detection of large-scale phenomena such as algal blooms and pollution spread [5]. However, these systems often lack high-resolution localized data and are affected by environmental factors such as cloud cover and atmospheric conditions.

Low-cost monitoring systems have also been proposed to make environmental monitoring more accessible. These systems focus on affordability and simplicity, using basic sensors and microcontrollers for data acquisition. Such solutions are particularly useful for small-scale applications like aquaculture and inland water monitoring [7]. However, their limited functionality restricts their application in complex marine environments.

2.2 Limitations of Existing Systems

Despite significant advancements in monitoring technologies, existing systems face several limitations that hinder their widespread adoption and effectiveness. One of the primary challenges is the dependence on manual intervention in traditional monitoring methods. These methods are not only time-consuming but also fail to provide continuous and real-time data, making them unsuitable for dynamic environmental conditions [1].

IoT-based systems, although more advanced, also have certain limitations. Many systems rely on centralized processing, which can lead to delays in data analysis and reduced system responsiveness. Additionally, communication technologies used in these systems may suffer from signal interference, limited range, and data loss, especially in harsh marine environments [4].

Remote sensing technologies, while useful for large-scale monitoring, have inherent limitations. These systems are highly dependent on environmental conditions such as cloud cover, atmospheric disturbances, and water surface reflections, which can affect data accuracy [5]. Moreover, remote sensing does not provide detailed localized information, which is essential for precise environmental analysis.

Another major limitation is the lack of adaptability in many buoy-based systems. Most existing solutions are designed for specific applications such as aquaculture or river monitoring, making them less suitable for diverse marine environments [6]. This lack of flexibility limits their usability in different environmental conditions.

Energy management is also a critical issue in existing systems. Many monitoring systems depend on external power sources or batteries, which require frequent maintenance and replacement. This increases operational costs and limits the duration of deployment, particularly in remote locations [2]. The absence of efficient energy harvesting techniques further restricts the sustainability of these systems.

Furthermore, many existing systems lack intelligent data processing capabilities. While they provide real-time data visualization, they do not effectively analyze trends or predict future environmental conditions. The absence of predictive analysis limits their ability to detect potential hazards such as pollution spikes or algal blooms in advance [4].

High implementation costs and complex system architectures also pose challenges for large-scale deployment. Advanced monitoring systems often require expensive sensors, communication modules, and infrastructure, making them less accessible for widespread use [3]. These limitations highlight the need for a more efficient, cost-effective, and intelligent monitoring solution.

2.3 Proposed System Overview

The proposed system aims to address the limitations of existing monitoring solutions by developing a self-sustaining solar-powered buoy for real-time environmental monitoring. The system integrates sensing, data processing, wireless communication, and cloud-based visualization into a single autonomous platform [2]. This integration enables efficient data acquisition, transmission, and analysis.

In the proposed system, multiple sensors are used to collect environmental and water quality data. The data is processed by an embedded microcontroller, which ensures proper formatting and synchronization before transmission.

A wireless communication module is used to transmit the data to a receiver unit, enabling seamless communication between different components of the system.

The receiver unit plays a crucial role in uploading the collected data to a cloud platform. This enables real-time monitoring and historical data analysis through a user-friendly interface. The cloud-based system allows users to access data remotely, improving accessibility and convenience.

A key feature of the proposed system is the integration of a machine learning model for predictive analysis. The model analyzes sensor data to identify patterns and detect anomalies, enabling early prediction of environmental hazards such as algal bloom formation [4]. This enhances the system's capability beyond simple monitoring, making it an intelligent environmental monitoring solution.

The system is powered by solar energy, ensuring continuous and maintenance-free operation. The use of renewable energy reduces dependence on external power sources and enhances the sustainability of the system. This makes the proposed solution suitable for deployment in remote and inaccessible aquatic environments.

Compared to existing systems, the proposed solution offers several advantages, including improved energy efficiency, real-time monitoring, predictive analysis, and cost-effectiveness. The system is designed to be scalable, allowing for future enhancements and large-scale deployment. Overall, the proposed system provides a comprehensive and sustainable solution for environmental monitoring and pollution analysis.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

3.1 Block Diagram

The proposed system consists of sensing, processing, communication, and cloud-based monitoring units integrated into a single architecture. Sensors collect environmental and water-related data, which is processed by the microcontroller. The processed data is transmitted wirelessly to a receiver unit using a communication module. The receiver uploads the data to a cloud platform for real-time monitoring and storage. A machine learning module analyzes the data for predictive insights. The system is powered by solar energy, ensuring continuous and autonomous operation.

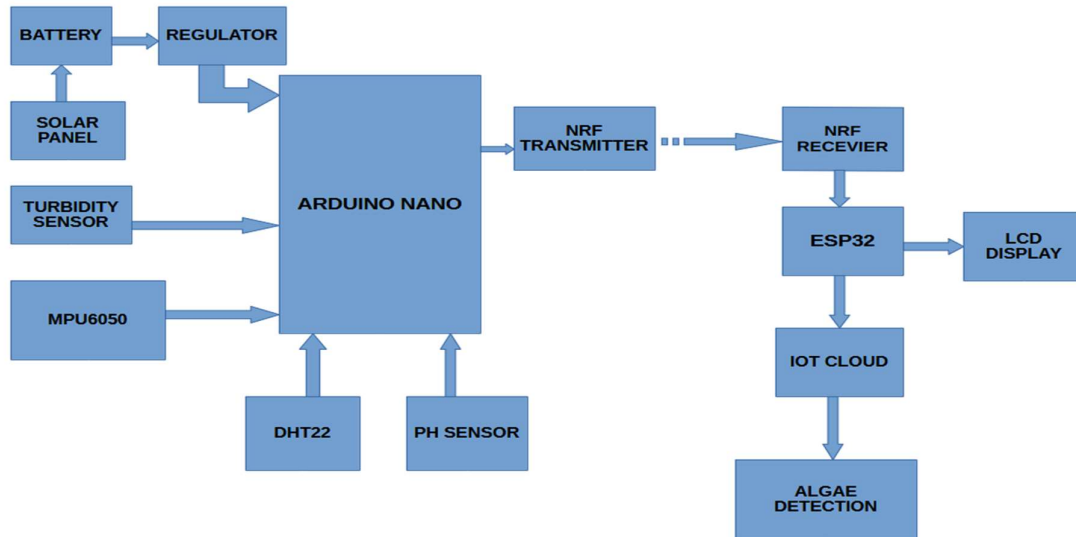


Fig 3.1: Block diagram of Self-Sustaining Solar Buoy for Oceanic Weather and Pollution Analysis

Solar Panel and Battery

The solar panel charges the battery, providing continuous power to the system. A voltage regulator ensures stable power supply to all components.

Arduino Nano

Acts as the main control unit, interfacing with sensors to collect and process data. It manages power usage and controls wireless transmission through the nRF24L01 module.

Sensors

- Turbidity Sensor: Measures water clarity and pollution level.
- pH Sensor: Monitors water acidity or alkalinity.
- DHT22: Records temperature and humidity.
- MPU6050: Detects motion and stability of the buoy.

nRF24L01 Transmitter and Receiver

Facilitates wireless data transfer between the buoy and the shore station, ensuring reliable long-range communication.

ESP32

Receives transmitted data, displays it on an LCD, and uploads it to the IoT cloud for remote monitoring and analysis.

IoT Cloud

Stores and visualizes the collected data, enabling real-time access and remote tracking of water quality parameters.

Algae Detection Module

Analyzes sensor data trends to detect early signs of **algae formation** based on changes in turbidity, pH, and temperature.

3.2 Working Principle

The proposed system operates by continuously monitoring environmental parameters using sensors. The collected data is processed by a microcontroller and transmitted wirelessly to a receiver unit, which uploads it to a cloud platform for real-time monitoring and visualization. A machine learning model analyzes the data to detect abnormal patterns and predict environmental hazards such as algae bloom formation. The system is powered by a solar panel with a rechargeable battery, ensuring uninterrupted operation in remote locations. This integration of sensing, communication, and intelligent analysis enables efficient and autonomous environmental monitoring with minimal human intervention.

3.3 Algorithm

The algorithm for the proposed system describes the step-by-step operation of data acquisition, processing, transmission, and analysis.

Step 1: Initialize the system components including sensors, microcontroller, and communication modules.

Step 2: Collect environmental data from the sensors.

Step 3: Process and format the collected data in the microcontroller.

Step 4: Transmit the processed data to the receiver unit using wireless communication.

Step 5: Upload the received data to the cloud platform.

Step 6: Store and display the data for real-time monitoring and historical analysis.

Step 7: Apply machine learning algorithms to analyze data trends and predict hazards.

Step 8: Generate alerts if abnormal conditions are detected.

Step 9: Repeat the process continuously for real-time monitoring

CHAPTER 4

HARDWARE COMPONENTS

The hardware components form the backbone of the proposed system. They are responsible for sensing environmental parameters, processing the data, transmitting it wirelessly, and ensuring continuous power supply.

4.1 Zero PCB

4.1.1 Role in the System

The Zero PCB acts as the fundamental hardware platform on which all electronic components of the transmitter unit are assembled and interconnected. It provides a structured base for soldering components such as microcontrollers, sensors, voltage regulators, and communication modules. The electrical connections between components are established through copper tracks, ensuring proper signal flow and stable circuit operation. In this system, the Zero PCB plays a critical role in maintaining circuit organization and preventing loose wiring issues that could otherwise lead to signal disturbances or system failure.

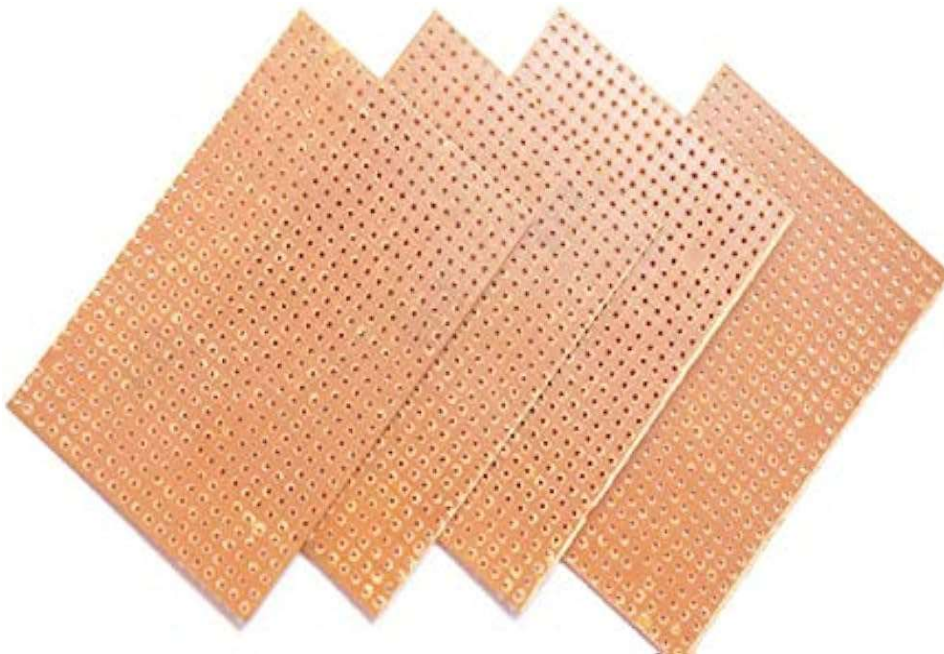


Fig. 4.1 Zero PCB Board

4.1.2 Justification for Selection

The Zero PCB has been selected for this project primarily because it offers high flexibility during the prototyping stage of development. Unlike custom-designed PCBs, which require fabrication and cannot be easily modified, the Zero PCB allows developers to quickly rearrange or replace components during testing and debugging phases. This flexibility is extremely important in academic projects where circuit optimization is an iterative process. Additionally, it eliminates the need for expensive PCB manufacturing, making it a cost-effective solution for prototype development.

4.1.3 Advantages

One of the major advantages of using Zero PCB is its ability to support rapid prototyping and experimentation. It significantly reduces wiring complexity and minimizes the chances of loose or faulty connections, which enhances the overall reliability of the system. Furthermore, it allows compact arrangement of components, which is essential in embedded systems where space optimization is critical. It is also durable enough to support long-term usage in experimental setups.

4.1.4 Applications

Zero PCB is widely used in electronics prototyping, embedded system development, robotics projects, IoT-based hardware implementations, and academic laboratory experiments. It is particularly useful in early-stage product development where circuit designs are frequently modified and tested.

4.1.5 Key Specifications

- Type: Prototype PCB (FR-2/FR-4 based)
- Hole Pitch: 2.54 mm standard grid
- Copper Tracks: Pre-defined horizontal/vertical pads (or none in strip type)
- Material: Glass epoxy (FR-4) or paper phenolic (FR-2)
- Operating Voltage: Not applicable
- Current Capacity: Depends on track width and soldering quality

4.2 Arduino Nano Microcontroller

4.2.1 Role in the System

The Arduino Nano serves as the central processing unit of the transmitter system and is responsible for coordinating the entire data acquisition and processing process. It continuously collects input signals from multiple sensors such as temperature, humidity, pH, turbidity, and motion sensors. After acquiring the raw data, it processes and converts it into meaningful digital values that can be transmitted or displayed. Additionally, it controls communication between different modules such as wireless RF transmitters and display units, ensuring smooth system operation.

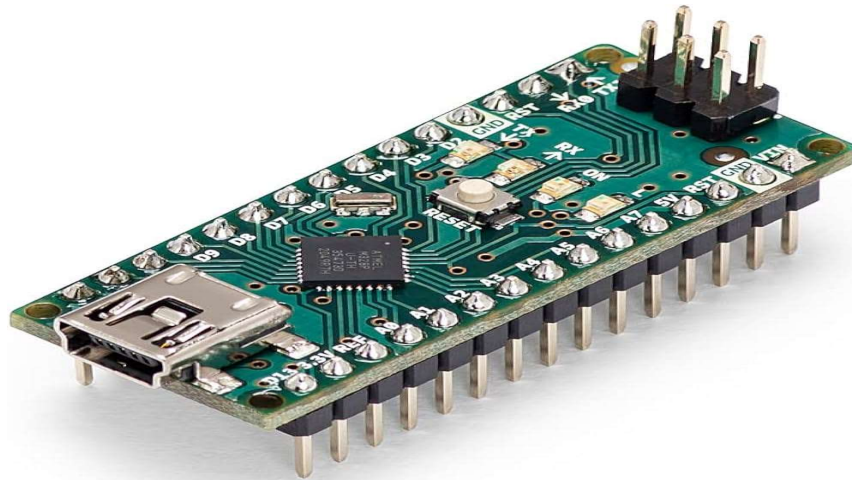


Fig. 4.2 Arduino Nano Microcontroller

4.2.2 Justification for Selection

The Arduino Nano has been chosen because it provides an ideal balance between performance, size, and cost. Since the system is intended for compact deployment in a buoy structure, space optimization is a major requirement, and the small form factor of Arduino Nano makes it highly suitable. It is also based on the ATmega328P microcontroller, which provides sufficient processing speed and memory for handling multiple sensor inputs simultaneously. Moreover, its compatibility with the Arduino IDE simplifies programming and reduces development complexity.

4.2.3 Advantages

The Arduino Nano offers several advantages including low power consumption, high processing efficiency, and ease of integration with a wide range of sensors and modules. It supports multiple digital and analog input/output pins, making it highly versatile for embedded applications. Additionally, it is supported by a large open-source community, which provides extensive libraries and documentation for rapid development.

4.2.4 Applications

Arduino Nano is widely used in IoT systems, smart automation projects, robotics control systems, environmental monitoring devices, and embedded sensor networks where compact and efficient microcontroller solutions are required.

4.2.5 Key Specifications

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Input Voltage (recommended): 7–12V
- Digital I/O Pins: 14 (6 PWM outputs)
- Analog Input Pins: 8
- Flash Memory: 32 KB
- SRAM: 2 KB
- EEPROM: 1 KB
- Clock Speed: 16 MHz

4.2.6 Pin Configuration

- Digital Pins: D0–D13 (GPIO, PWM, UART, SPI)
- Analog Pins: A0–A7 (sensor inputs)
- Power Pins: 5V, 3.3V, GND, VIN
- Communication Pins:
 - TX (D1), RX (D0) → Serial communication
 - SDA (A4), SCL (A5) → I2C communication

4.3 LM7805 Voltage Regulator (5V)

4.3.1 Role in the System

The LM7805 voltage regulator is responsible for maintaining a constant 5V power supply for the microcontroller and other electronic components in the system. Since the system is powered by a solar panel and rechargeable battery, the input voltage is not stable and tends to fluctuate depending on environmental conditions. The LM7805 ensures that these fluctuations do not affect the sensitive electronic components by providing a steady and regulated output voltage.



Fig. 4.3 LM7805 Voltage Regulator

4.3.2 Justification for Selection

The LM7805 has been selected because of its simplicity, reliability, and proven performance in electronic power supply circuits. It does not require complex external circuitry and can easily be integrated into the system. Its ability to handle varying input voltages while providing a stable output makes it ideal for renewable energy-based systems where power inconsistency is common.

4.3.3 Advantages

The LM7805 offers stable output voltage regulation, built-in thermal shutdown protection, and short-circuit protection features. It is highly reliable for low-power electronic applications and ensures long-term stability of the system without frequent maintenance requirements.

4.3.4 Applications

It is commonly used in embedded systems, regulated power supply circuits, battery-powered devices, and microcontroller-based electronic systems.

4.3.5 Key Specifications

- Output Voltage: 5V fixed
- Input Voltage Range: 7V – 35V
- Output Current: Up to 1A (with heat sink)
- Type: Linear regulator
- Protection: Thermal shutdown, short circuit protection

4.3.6 Pin Configuration

- Pin 1: Input (V_{in})
- Pin 2: Ground (GND)
- Pin 3: Output ($V_{out} = 5V$)

4.4 LM7833 3.3V Voltage Regulator

4.4.1 Role in the System

The LM7833 3.3V voltage regulator is used in the system to provide a stable and regulated 3.3V power supply to low-voltage electronic components. In the proposed solar buoy system, certain modules such as the RF transceiver and other communication or sensor interfaces require a precise 3.3V input for proper operation. Since the input voltage obtained from the solar panel and rechargeable battery is not constant and may fluctuate due to environmental conditions, the LM7833 ensures that a steady output voltage is maintained. This helps in protecting sensitive components from voltage variations and ensures reliable system performance.

4.4.2 Justification for Selection

The LM7833 voltage regulator is selected because it is specifically designed to provide a fixed 3.3V output, which is essential for modern low-power electronic devices. Many communication modules and sensors operate only at 3.3V, and supplying higher voltage can damage these components. The LM7833 provides a simple and efficient solution for stepping down higher input voltages to a safe operating level. It is also easy to implement, cost-effective, and suitable for embedded systems where stable voltage regulation is required without complex circuitry.

4.4.3 Advantages

The LM7833 regulator offers several advantages that make it suitable for use in this system. It provides a constant and stable output voltage, ensuring reliable operation of connected components. It has built-in protection features such as thermal shutdown and short-circuit protection, which enhance system safety. The device is easy to use and requires minimal external components, making it ideal for compact circuit designs. Additionally, it offers good performance with low noise output, which is important for communication modules and sensor accuracy.



Fig. 4.4 LM7833 3.3V Voltage Regulator

4.4.4 Applications

The LM7833 3.3V voltage regulator is widely used in various electronic applications where a stable low-voltage supply is required. It is commonly used in IoT systems, wireless communication devices, microcontroller-based circuits, and sensor modules. It is also used in portable electronics, battery-powered devices, and embedded systems where efficient and reliable voltage regulation is necessary.

4.4.5 Key Specifications

- Output Voltage: 3.3V (fixed)
- Input Voltage Range: 5V – 20V (typical operating range)
- Output Current: Up to 1A (with proper heat dissipation)
- Type: Linear Voltage Regulator
- Protection Features: Thermal shutdown, short circuit protection
- Output Regulation: Stable with low ripple

4.4.6 Pin Configuration

- **Pin 1:** Input Voltage (V_{in})
- **Pin 2:** Ground (GND)
- **Pin 3:** Output Voltage (3.3V)

4.5 nRF24L01 RF Transceiver Module

4.5.1 Role in the System

The nRF24L01 RF transceiver module is responsible for establishing wireless communication between the transmitter unit and the receiver unit. It transmits sensor data collected from the buoy to the receiving system using radio frequency communication in the 2.4 GHz ISM band. This enables real-time remote monitoring of environmental conditions without the need for physical connections.

4.5.2 Justification for Selection

The module has been selected due to its excellent balance between power efficiency, data transmission speed, and communication reliability. Since the system is powered by solar energy, low power consumption is a critical requirement, and the nRF24L01 meets this requirement effectively. It also supports multiple channels and provides stable communication over moderate distances.



Fig. 4.5 nRF24L01 RF Transceiver Module

4.5.3 Advantages

It offers low energy consumption, high data transmission speed, multi-channel support, and reliable wireless connectivity. It is also cost-effective compared to other wireless communication technologies.

4.5.4 Applications

It is widely used in wireless sensor networks, IoT communication systems, remote monitoring applications, and robotics communication systems.

4.5.5 Key Specifications

- Operating Voltage: 1.9V – 3.6V (3.3V typical)
- Communication: SPI interface
- Frequency: 2.4 GHz ISM band
- Data Rate: 250 kbps, 1 Mbps, 2 Mbps
- Range: Up to 100 meters (open space)
- Power Consumption: Very low (sleep mode supported)

4.5.6 Pin Configuration

- VCC: 3.3V supply
- GND: Ground
- CE: Chip Enable
- CSN: Chip Select Not (SPI control)
- SCK: SPI Clock
- MOSI: Master Out Slave In
- MISO: Master In Slave Out

4.6 Solar Panel

4.6.1 Role in the System

The solar panel is one of the most important components in the proposed system as it is responsible for converting sunlight into electrical energy through the photovoltaic effect. This generated electrical energy is used to power the entire buoy system and also charge the rechargeable battery for continuous operation. Since the system is designed to operate in remote aquatic environments where conventional electricity supply is not available, the solar panel ensures complete energy independence. It continuously harvests solar energy during daytime and supports uninterrupted system functionality by supplying power to sensors, microcontrollers, and communication modules.

4.6.2 Justification for Selection

The solar panel has been selected because the system is intended for long-term deployment in open water environments where maintenance and external power supply are difficult. Solar energy is the most reliable and sustainable source of power for such applications. It reduces dependency on grid electricity and eliminates the need for frequent battery replacement. Additionally, solar panels are environmentally friendly and align with the objective of developing a self-sustaining monitoring system.



Fig. 4.6 Solar Panel

4.6.3 Advantages

The major advantage of using a solar panel is its ability to provide renewable and clean energy without producing any pollution. It significantly reduces operational costs and ensures continuous energy availability in remote areas. Solar panels also have long operational lifespans, low maintenance requirements, and high reliability when exposed to sufficient sunlight.

4.6.4 Applications

Solar panels are widely used in renewable energy systems, remote monitoring stations, satellites, IoT-based field devices, agricultural automation systems, and marine monitoring systems where grid electricity is not accessible.

4.6.5 Key Specifications

- Type: Monocrystalline / Polycrystalline
- Output Voltage: 6V / 12V (depending on design)
- Power Rating: 1W – 20W (project dependent)
- Efficiency: 15% – 22%
- Lifespan: 20–25 years

4.7 Rechargeable Battery

4.7.1 Role in the System

The rechargeable battery acts as an energy storage unit in the system. It stores excess electrical energy generated by the solar panel during daytime and supplies power to the system during nighttime or periods of low sunlight. This ensures that the system operates continuously without interruption, even when solar energy is not available. It provides stable power to all components including sensors, microcontrollers, and communication modules.

4.7.2 Justification for Selection

The rechargeable battery is selected because energy storage is essential for maintaining continuous operation of a solar-powered system. Without a battery, the system would only function during sunlight hours, which is not practical for real-time monitoring applications. A rechargeable battery ensures energy availability at all times and supports long-term autonomous operation of the buoy system.

4.7.3 Advantages

Rechargeable batteries provide reliable energy storage, long lifecycle with multiple charge-discharge cycles, and stable output voltage. They are reusable, cost-effective in the long run, and compatible with solar charging systems. They also help in maintaining system stability during sudden power fluctuations.

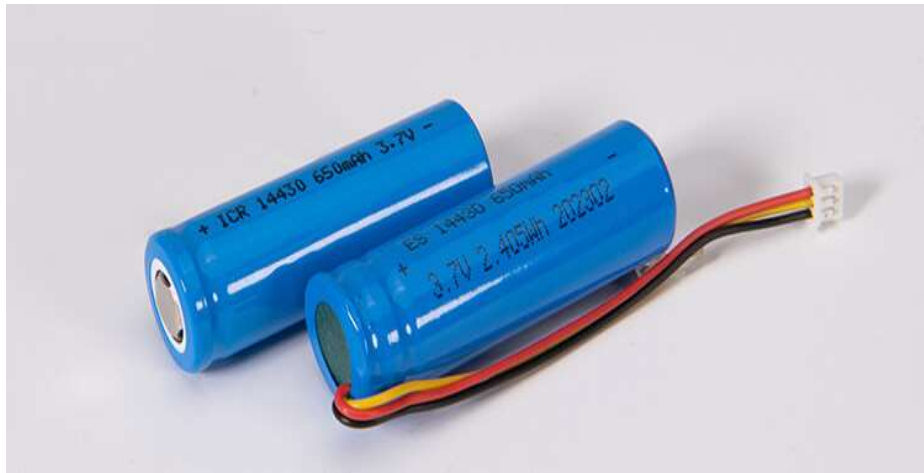


Fig. 4.7 Rechargeable Battery

4.7.4 Applications

Rechargeable batteries are widely used in solar energy systems, electric vehicles, portable electronic devices, backup power systems, and remote IoT monitoring devices.

4.8 Battery Management System (BMS)

4.8.1 Role in the System

The Battery Management System (BMS) is responsible for monitoring and controlling the charging and discharging process of the rechargeable battery. It ensures that the battery operates within safe voltage and current limits.

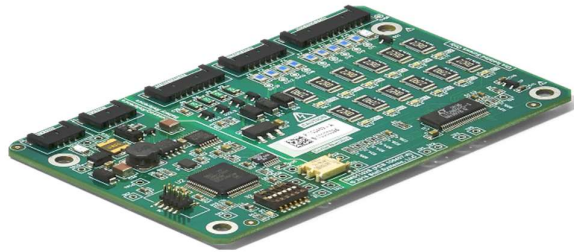


Fig. 4.8 Battery Management System

4.8.2 Justification for Selection

The BMS is selected because safety and reliability are critical in autonomous solar-powered systems.

4.8.3 Advantages

It enhances battery safety, extends battery lifespan, and ensures stable energy delivery.

4.8.4 Applications

Battery Management Systems are widely used in electric vehicles, solar power storage systems, drones, portable electronics, and large-scale energy storage solutions.

4.8.5 Key Specifications

- Voltage Range: 3.7V / 7.4V / 12V support
- Protection: Overcharge, over-discharge, short circuit
- Balancing: Cell balancing support
- Current Rating: 5A – 20A (varies by module)

4.8.6 Pin Configuration

- B+ : Battery Positive
- B- : Battery Negative
- P+ : Load/Output Positive
- P- : Load/Output Negative

4.9 MPU6050 Sensor Module

4.9.1 Role in the System

The MPU6050 sensor module is used to measure motion, acceleration, and orientation of the buoy system. It combines both an accelerometer and gyroscope in a single module, allowing precise detection of tilt, vibration, and movement. This data is essential for monitoring the stability and behavior of the buoy in water conditions such as waves and currents.



Fig. 4.9 MPU6050 Sensor Module

4.9.2 Justification for Selection

The MPU6050 is selected because it provides accurate motion sensing in a compact and cost-effective module. It eliminates the need for separate accelerometer and gyroscope sensors by integrating both functionalities into a single chip, reducing system complexity and improving efficiency.

4.9.3 Advantages

It provides high accuracy motion detection, low power consumption, compact design, and easy interfacing with microcontrollers. It also offers stable performance in dynamic environmental conditions.

4.9.4 Applications

The MPU6050 is widely used in robotics, drones, motion tracking systems, navigation systems, gaming controllers, and stabilization systems.

4.9.5 Key Specifications

- Operating Voltage: 3.3V – 5V
- Sensor Type: 3-axis accelerometer
- Communication: I2C
- Range:
 - Gyroscope: ± 250 , ± 500 , ± 1000 , ± 2000 °/s
- Resolution: 16-bit

4.9.6 Pin Configuration

- VCC: Power supply
- GND: Ground
- SDA: Data line (I2C)
- SCL: Clock line (I2C)
- INT: Interrupt pin (optional)

4.10 DHT22 Temperature and Humidity Sensor

4.10.1 Role in the System

The DHT22 sensor is used to measure atmospheric temperature and humidity in the surrounding environment of the buoy. This environmental data is important for analyzing weather conditions and understanding climatic variations in the monitored area.



Fig. 4.10 DHT22 Sensor

4.10.2 Justification for Selection

The DHT22 is selected because it provides higher accuracy and better long-term stability compared to similar low-cost sensors. It is suitable for outdoor environmental monitoring applications where reliable and consistent readings are required.

4.10.3 Advantages

It offers high measurement accuracy, stable performance, low power consumption, and simple digital output interfacing. It also has good resistance to environmental variations.

4.10.4 Applications

The DHT22 sensor is used in weather monitoring systems, smart agriculture, greenhouse control systems, HVAC systems, and environmental data collection systems.

4.10.5 Key Specifications

- Operating Voltage: 3.3V – 5V
- Temperature Range: -40°C to +80°C
- Humidity Range: 0% – 100%
- Accuracy:
 - Temperature: $\pm 0.5^{\circ}\text{C}$
 - Humidity: $\pm 2\text{--}5\%$
- Sampling Rate: 0.5 Hz (2 sec interval)

4.10.6 Pin Configuration

- VCC: Power supply
- DATA: Digital output
- GND: Ground

4.11 pH Sensor Module

4.11.1 Role in the System

The pH sensor module is used to measure the acidity or alkalinity level of water in the monitoring system. This parameter is extremely important in determining water quality and detecting possible pollution in aquatic environments. The sensor continuously monitors pH levels and provides real-time data for analysis.

4.11.2 Justification for Selection

The pH sensor is selected because water quality assessment is a major objective of the proposed system. Monitoring pH levels helps in identifying chemical imbalances and pollution levels in water bodies. This makes it essential for environmental protection and research applications.



Fig. 4.11 pH Sensor Module

4.11.3 Advantages

It provides accurate measurement of hydrogen ion concentration, real-time monitoring capability, and easy integration with microcontrollers. It is also suitable for continuous water quality analysis.

4.11.4 Applications

pH sensors are widely used in water treatment plants, aquaculture systems, environmental monitoring projects, chemical industries, and laboratory testing.

4.11.5 Key Specifications

- Operating Voltage: 5V
- Measurement Range: 0 – 14 pH
- Accuracy: ± 0.1 pH (approx.)
- Response Time: Fast real-time response
- Output: Analog voltage

4.11.6 Pin Configuration

- VCC: Power supply
- GND: Ground
- Analog Output (AO): pH signal output

4.12 Turbidity Sensor Module

4.12.1 Role in the System

The turbidity sensor measures the clarity of water by detecting the amount of suspended particles present in it. This helps in determining the level of contamination or pollution in water bodies. In the proposed system, it plays a crucial role in assessing real-time water quality.



Fig. 4.12 Turbidity Sensor Module

4.12.2 Justification for Selection

The turbidity sensor is selected because visual water clarity is a key indicator of pollution and sediment levels. It provides a direct measurement of water quality and helps in detecting environmental changes at an early stage.

4.12.3 Advantages

It offers real-time monitoring, high sensitivity to suspended particles, simple operation, and reliable output readings. It is also easy to integrate with embedded systems.

4.12.4 Applications

Turbidity sensors are used in water treatment plants, river and lake monitoring systems, environmental research, and pollution detection systems.

4.12.5 Key Specifications

- Operating Voltage: 5V
- Measurement Range: 0 – 3000 NTU (approx.)
- Output Type: Analog
- Response Time: Real-time detection
- Principle: Light scattering

4.12.6 Pin Configuration

- VCC: Power supply
- GND: Ground
- Analog Output (AO): Turbidity value

4.13 ESP32 Microcontroller

4.13.1 Role in the System

The ESP32 microcontroller acts as the receiver and cloud communication unit in the system. It receives processed sensor data from the transmitter unit and uploads it to the cloud using built-in Wi-Fi connectivity. It enables real-time remote monitoring and data visualization through IoT platforms.

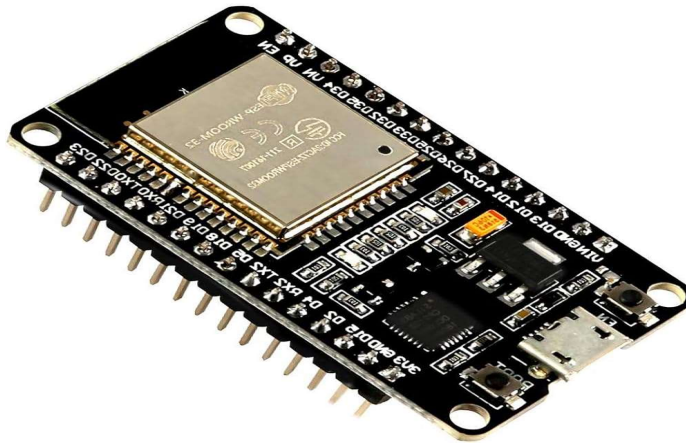


Fig. 4.13 ESP32 Microcontroller

4.13.2 Justification for Selection

The ESP32 is selected because it combines high processing power with built-in wireless communication features. It eliminates the need for external Wi-Fi modules and provides a complete IoT solution in a single chip.

4.13.3 Advantages

It offers dual-core processing, Wi-Fi and Bluetooth support, low power consumption modes, and strong cloud integration capability. It is highly efficient for IoT-based applications.

4.13.4 Applications

ESP32 is widely used in IoT systems, smart home automation, industrial monitoring, cloud-connected devices, and real-time data logging systems.

4.13.5 Key Specifications

- Processor: Dual-core Tensilica LX6
- Operating Voltage: 3.3V
- Wi-Fi: 802.11 b/g/n
- Bluetooth: v4.2 (BLE + Classic)
- GPIO Pins: 30+
- ADC Channels: 12-bit

4.13.6 Pin Configuration

- 3V3: Power supply
- GND: Ground
- EN: Enable pin
- GPIO Pins: General purpose I/O
- TX/RX: UART communication
- SDA/SCL: I2C communication

4.14 LCD Display

4.14.1 Role in the System

The LCD display is used to show real-time sensor readings locally within the system. It provides immediate visual feedback of environmental parameters such as temperature, humidity, pH, and turbidity without requiring internet access.



Fig. 4.14 LCD Display

4.14.2 Justification for Selection

The LCD is selected because it allows quick on-site monitoring of system data, which is useful during testing, calibration, and maintenance phases. It ensures that users can view system status even without cloud connectivity.

4.14.3 Advantages

It provides clear visual output, low power consumption, easy interfacing, and real-time display capability. It is reliable and widely used in embedded systems.

4.14.4 Applications

LCD displays are used in embedded systems, industrial monitoring panels, IoT devices, and electronic measurement instruments.

4.14.5 Key Specifications

- Operating Voltage: 5V
- Display Format: 16 characters × 2 lines
- Controller: HD44780
- Backlight: LED
- Interface Mode: 4-bit / 8-bit parallel

4.14.6 Pin Configuration

- VSS: Ground
- VDD: Power supply
- V0: Contrast adjustment
- RS: Register Select
- RW: Read/Write
- E: Enable pin
- D0–D7: Data pins
- A/K: Backlight power

CHAPTER 5

SOFTWARE REQUIREMENTS

5.1 Arduino IDE

Arduino IDE is used for writing, compiling, and uploading code to the microcontrollers. It enables sensor data acquisition, processing, and communication between hardware components.

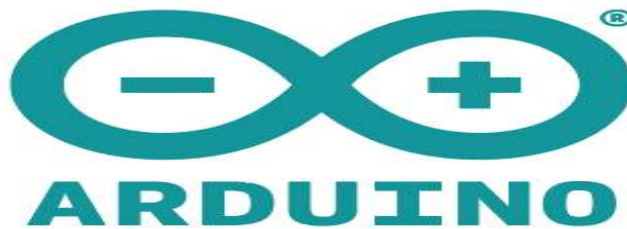


Fig. 5.1 Arduino IDE Interface

5.2 Blynk IoT Platform

Blynk IoT platform is used for real-time data monitoring and visualization. It provides a user-friendly interface to display sensor data and track system performance remotely.



Fig. 5.2 Blynk Dashboard Interface

5.3 Google Colaboratory

Google Colaboratory is used for implementing and running machine learning models. It allows data analysis, prediction of environmental conditions, and development of algorithms using Python.



Fig. 5.3 Google Colaboratory

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Observations

The developed self-sustaining solar buoy system was successfully implemented and tested for real-time environmental monitoring. The prototype integrates transmitter and receiver units along with a solar-powered energy system. It effectively acquires sensor data, processes it using the microcontroller, and transmits it wirelessly with minimal data loss. The system operated consistently under different testing conditions, demonstrating stable communication and reliable performance over extended durations.



Fig. 6.1 Final Prototype of Water Buoy System

The data received at the receiver unit was successfully uploaded to the cloud platform for visualization. The system provided continuous and real-time updates, allowing users to monitor environmental parameters remotely through an intuitive interface. The displayed data was clear, responsive, and accurately reflected the sensed conditions. The system maintained proper synchronization between hardware and software components, and no major interruptions were observed during operation, indicating dependable system behavior.



Fig. 6.2 Blynk Dashboard Output

6.2 Performance Analysis

The performance of the proposed system was evaluated based on reliability, efficiency, and responsiveness. The wireless communication between the transmitter and receiver units was found to be stable, ensuring continuous data transmission over the required range. The system exhibited minimal latency, making it suitable for real-time monitoring applications.

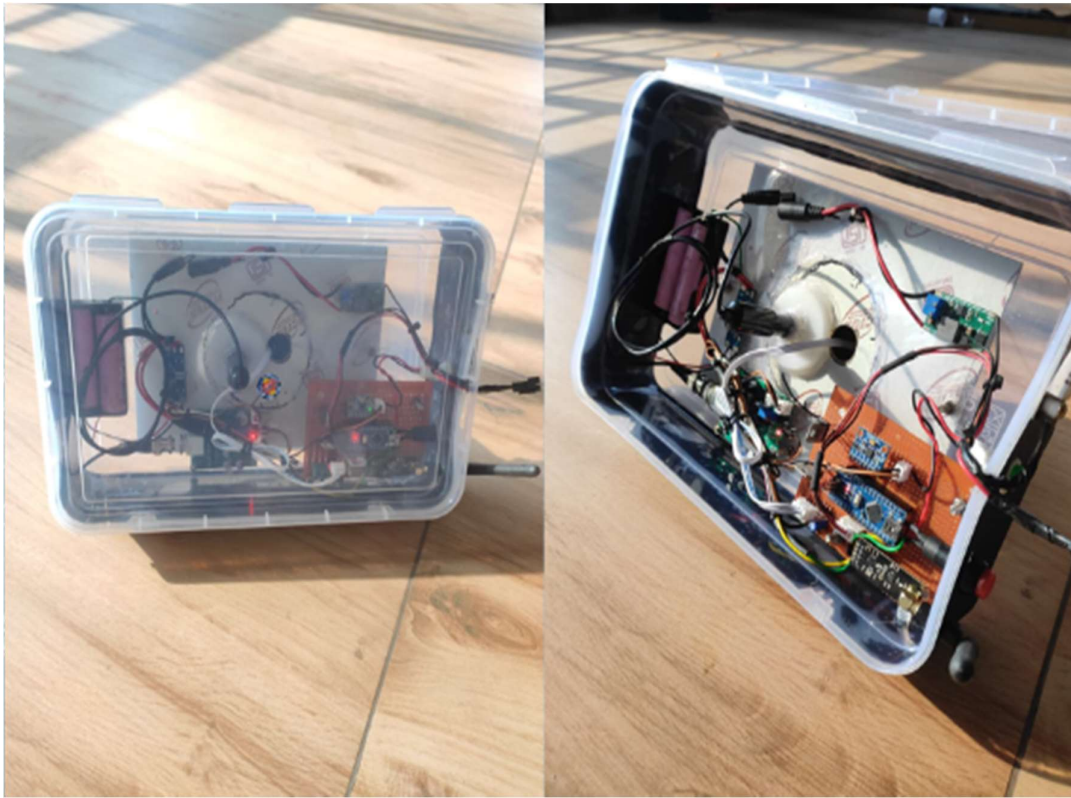


Fig. 6.3 Transmitter Unit Setup

The integration of sensors and microcontroller ensured accurate and consistent data acquisition, contributing to the overall reliability of the system. The use of solar energy improved system efficiency by enabling continuous operation without dependence on external power sources, even in remote environments. The machine learning model enhanced system capability by providing predictive analysis based on collected data. Overall, the system demonstrated good accuracy, energy efficiency, scalability, and robustness, making it suitable for long-term environmental monitoring applications.

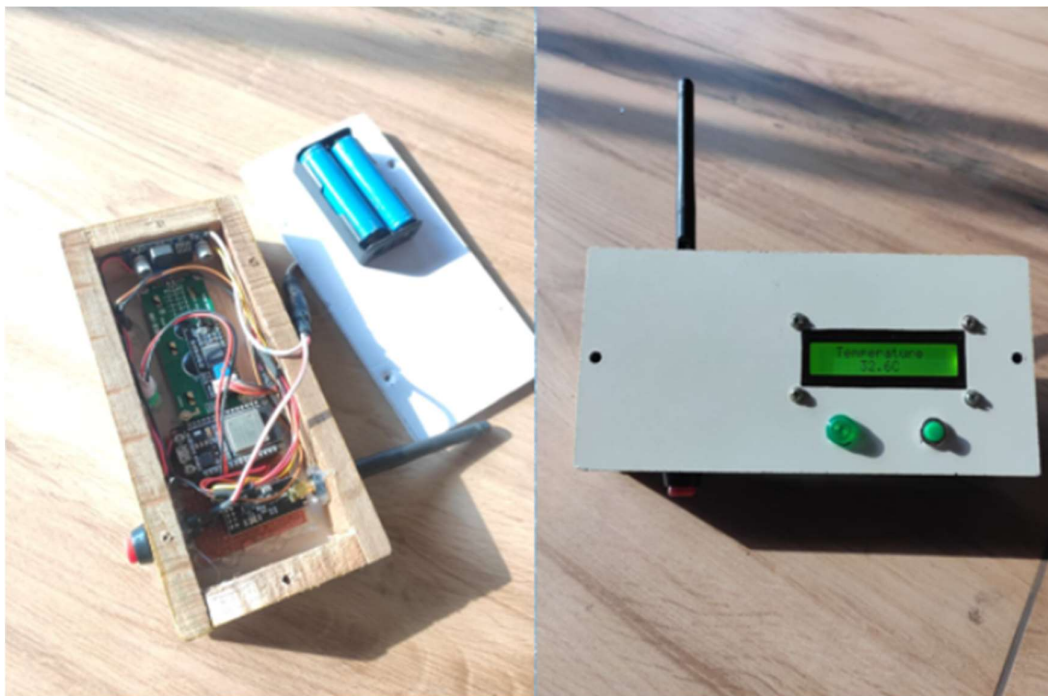


Fig. 6.4 Receiver Unit Setup

6.3 Advantages

- Provides real-time monitoring of environmental parameters, reducing the need for manual data collection.
- Enables remote access to data through wireless communication, improving convenience and efficiency.
- Ensures continuous operation using solar power, making the system self-sustaining.
- Requires minimal maintenance due to autonomous operation and renewable energy usage.
- Incorporates machine learning for predictive analysis of environmental conditions.
- Allows early detection of potential hazards such as pollution or algal blooms.
- Cost-effective solution compared to traditional monitoring systems.
- Scalable design suitable for deployment in oceans, lakes, and rivers.
- Environmentally friendly system due to the use of renewable energy sources.
- Provides reliable and continuous data for long-term environmental monitoring.

6.4 Discussion of Findings

The implementation of the self-sustaining solar buoy demonstrated that an integrated IoT and machine learning approach is highly effective for real-time environmental monitoring. The system successfully collected and transmitted data across various parameters, including water turbidity, pH, temperature, and humidity.

- **System Autonomy:** A primary success of this project was its energy independence. By utilizing solar panels and a Battery Management System (BMS), the buoy maintained continuous operation without requiring external power or manual battery replacement. This confirms the system's suitability for remote deployment where maintenance is difficult.

- **Data Accuracy and Transmission:** The use of the Arduino Nano and NRF24L01 modules facilitated reliable wireless data transfer between the buoy (transmitter) and the shore station (receiver). The integration with the Blynk IoT platform allowed for seamless real-time visualization, which is a significant improvement over traditional manual sampling methods that lack real-time capabilities.
- **Predictive Capabilities:** The inclusion of a Python-based machine learning model adds a layer of intelligence that many existing systems lack. By analyzing trends in pH, turbidity, and temperature, the system can provide early warnings for hazards like algal blooms, allowing for proactive environmental management.

6.5 Performance Analysis and Challenges

While the prototype performed reliably in testing, several observations were made regarding its operation in aquatic environments:

- **Sensor Sensitivity:** The pH and turbidity sensors provided critical water quality data, but their long-term accuracy in harsh marine conditions can be affected by biofouling (accumulation of microorganisms). This highlights the need for specialized sensor housings in future iterations.
- **Stability:** The MPU6050 sensor played a vital role in monitoring the buoy's stability against waves. Data from this sensor suggests that the physical housing design is crucial for ensuring the buoy remains upright to maximize solar intake.
- **Communication Range:** The NRF24L01 module provided stable communication over moderate distances. For deep-sea applications, the discussion must consider upgrading to LoRa or satellite communication to extend this range.

6.6 Environmental and Economic Impact

The proposed system addresses the high operational costs associated with conventional water quality monitoring.

- **Cost-Effectiveness:** By using affordable components like the ESP32 and Arduino Nano, the system offers a scalable solution that can be deployed in large numbers for comprehensive coverage of lakes and rivers.
- **Sustainability:** The self-powered design aligns with global environmental goals by reducing the carbon footprint of monitoring activities and eliminating the risk of battery leakage in sensitive aquatic ecosystems.

CHAPTER 7

APPLICATIONS

The proposed system can be applied in various environmental monitoring and marine-related fields due to its real-time data acquisition and analysis capabilities. Some of the key applications are listed below:

- Real-time monitoring of water quality in oceans, lakes, and rivers.
- Detection and analysis of water pollution levels.
- Monitoring environmental conditions in aquaculture and fish farming.
- Weather monitoring in coastal and marine regions.
- Useful for disaster management and early warning systems.
- Deployment in remote areas for continuous environmental observation.
- Can be used in research and marine studies.
- Integration with smart city and IoT-based environmental systems.

CHAPTER 8

FUTURE SCOPE

The proposed system can be further enhanced by incorporating advanced technologies and additional features to improve its performance and scalability. The possible future improvements are as follows:

- Integration of additional sensors for enhanced monitoring accuracy.
- Use of advanced communication technologies like LoRa or GSM for long-range transmission.
- Improvement of machine learning models using larger datasets.
- Addition of GPS module for location tracking of the buoy.
- Development of a mobile application for better user interaction.
- Enhancement of system design for large-scale deployment.
- Implementation of advanced data analytics for better prediction.
- Improvement in power management for higher efficiency.

CHAPTER 9

CONCLUSION

The project successfully demonstrates the design and implementation of a self-sustaining solar buoy system for real-time environmental monitoring. It effectively integrates various sensors, microcontrollers, wireless communication modules, and cloud-based IoT platforms to continuously collect, process, and display environmental data. This integration ensures accurate and reliable monitoring of key parameters such as water quality, temperature, humidity, and motion, enabling effective environmental analysis.

The use of solar energy makes the system completely autonomous and suitable for deployment in remote and inaccessible locations such as lakes, rivers, and coastal areas. By eliminating dependency on conventional power sources, the system ensures continuous operation with minimal maintenance. The incorporation of a rechargeable battery further enhances reliability by providing uninterrupted power during low sunlight conditions.

Additionally, the inclusion of machine learning techniques enhances the system's capability by enabling predictive analysis of environmental conditions based on collected data trends. Overall, the proposed system is cost-effective, efficient, and scalable, offering a sustainable solution for environmental and pollution monitoring. The project effectively highlights the potential of combining IoT, renewable energy, and intelligent data processing for advanced environmental monitoring applications.

REFERENCES

- [1] H. Li and M. M. Shafieezadeh, "Smart sensor architecture selection for coastal marine monitoring," *Applied Water Science*, vol. 15, no. 12, Art. no. 302, Nov. 2025. [Online]
- [2] Hoang -Yang Lu, Chih-Yung Cheng, Shyi-Chyi Cheng, Yu-Hao Cheng, Wen-Chen Lo, Wei-Lin Jiang, Fan-Hua Nan, Shun-Hsyung Chang, and Naomi A. Ubina, "A Low-Cost AI Buoy System for Monitoring Water Quality at Offshore Aquaculture Cages," *Sensors*, vol. 22, no. 11, Art. no. 4078, May 2022. [Online].
- [3] M. Amani *et al.*, "Ocean Remote Sensing Techniques and Applications: A Review (Part I)," *Water*, vol. 14, no. 21, Art. no. 3400, Oct. 2022. [Online]
- [4] A. Shukla, P. S. Matharu, and B. Bhattacharya, "Design and development of a continuous water quality monitoring buoy for health monitoring of river Ganga," *Engineering Research Express*, vol. 5, no. 4, Art. no. 045073, Nov. 2023. [Online]
- [5] H.-Y. Lu *et al.*, "A Low-Cost AI Buoy System for Monitoring Water Quality at Offshore Aquaculture Cages," *Sensors*, vol. 22, no. 11, Art. no. 4078, May 2022. [Online]
- [6] J.-W. Bian and C.-J. Huang, "Real-Time and Long-Term Monitoring of Coastal Water Turbidity Using an Ocean Buoy Equipped with an ADCP," *Sensors*, vol. 24, no. 21, Art. no. 6979, Oct. 2024. [Online].
- [7] S. F. Nasution, H. Harmadi, S. Suryadi, and B. Widiyatmoko, "Development of River Flow and Water Quality Using IOT-based Smart Buoys Environment Monitoring System," *Jurnal Ilmu Fisika*, vol. 16, no. 1, pp. 1–12, Mar. 2024. [Online]
- [8] J. D. Medina *et al.*, "Open-source low-cost design of a buoy for remote water quality monitoring in fish farming," *PLOS ONE*, vol. 17, no. 6, Art. no. e0270202, Jun. 2022. [Online]

